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# Chemical Bonding & Hybridisation — D4 Issue #38

## Candidate Specimen: Construction

**Contents and concept route**

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**Sections+**

**Conformational torsion and p-orbital alignment**

**Subject:** Chemistry

**Topic:** Chemical Bonding and Molecular Structure

**Concept:** Conformational torsion and p-orbital alignment

**Concept difficulty:** Hard concept

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## Construct both arrangements while preserving connectivity and electron inventories.

Hard concept

Section 1 of 10

Next →

### Conditions for this construction

- Understands sigma and pi bonding as head-on and lateral orbital overlaps.
- Can recognize parallel versus perpendicular geometric vectors in 3D.

#### Why this section matters for the question set

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#### 1 Inventory formamide electrons and connectivity.

**Why valid:**  $\text{CH}_3\text{NO}$  supplies  $4+3+5+6$  valence electrons and  $6+3+7+8$  total electrons.

**Result:** 18 valence; 24 total;  $\text{H}-\text{C}(=\text{O})-\text{NH}_2$  connectivity fixed.

#### 2 The step the hard question turns on Construct the supplied aligned donor/carbonyl pi account.

**Why valid:** The localized donor pair plus carbonyl pi pair gives four electrons in a three-centre qualitative pi space.

**Result:** Aligned N p-like donor and C/O p directions; four model pi-space electrons.

**3 Rotate the N donor direction about C-N toward 90 degrees.**

**Why valid:** Relative torsion changes orientation against the fixed carbonyl reference; it does not change element counts.

**Result:** Approximately orthogonal donor/acceptor directions with unchanged connectivity and electron ledger.

**4 Check what the two sketches can establish.**

**Why valid:** Orientation distinguishes alignment, while donation strength needs an experiment or specified calculation.

**Result:** Both drawings retain 18 valence/24 total electrons; no measured donation strength is claimed.

Fixed connectivity:  $\text{H-C(=O)-NH}_2$  in bc  
End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

**Keep alignment separate from total energy**

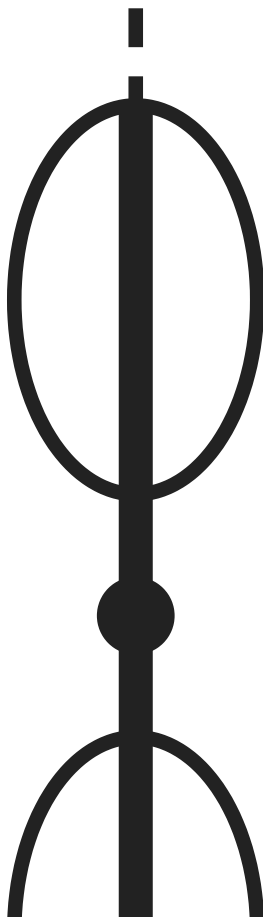
Look along the fixed reference axis. The two centres project to the same origin. The solid test direction turns relative to the dashed reference direction. This is a direction diagram, not a measured field or coupling integral.

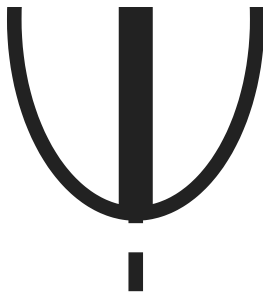
Relative angle  $\theta$  (degrees)



Independent toy energy contribution  $c$







# Axis points towards/away from viewer.

$\theta = 0^\circ$ ; given factor = 1; toy  $E = -1$ .

Declared demonstration only:  $g = \cos(\theta)$ ;  $E = -g^2 + c$  in arbitrary toy units. The square is an explicitly chosen example, not a universal law. Hold  $\theta$  fixed and vary  $c$ : the same  $g$  permits different  $E$ . Predict this before moving  $c$ . No physical barrier is computed.

## The key step

Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.

**Worked explanation**

Count valence and total electrons for CH<sub>3</sub>NO before and after torsion. Identify which geometric relation changes.

**Follow the operation, its justification and its result.**

1. Torsion changes relative orientation, not the number of atoms or formal electrons.
2. 18 valence, 24 total in both; relative donor direction changes.
3. Formamide CH<sub>3</sub>NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl pi model has four electrons. A 90-degree twist changes alignment, not these inventories.

**Result**

18 valence, 24 total in both; relative donor direction changes.

**Check**

Independent element sums: valence  $4+3+5+6=18$ ; total  $6+3+7+8=24$ . Both sketches must preserve them.

**Mistake clinic • diagnose and repair**

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Independent element sums: valence  $4+3+5+6=18$ ; total  $6+3+7+8=24$ . Both sketches must preserve them.

#### 2 • APPLY

Sketch a fixed-connectivity 45-degree comparison without claiming measured donation.

#### 3 • CONNECT

Torsion changes relative orientation, not the number of atoms or formal electrons.

**2 Explain the supplied trend without replacing a mechanism with a label.**

**Hard concept**

Section 2 of 10

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**Why this section matters for the question set**

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**1 Use the supplied model result as a conditional premise.**

**Why valid:** The benchmark specifies reduced alignment and delocalisation contribution toward 90-degree twist.

**Result:** Connectivity fixed; relevant orbital directions rotate relative to each other.

**2 The step the hard question turns on Connect torsion to the overlap mechanism.**

**Why valid:** Parallel p-like donor/acceptor directions support lateral interaction; orthogonal directions reduce that particular alignment.

**Result:** Reduced alignment explains the stated qualitative contribution change.

**3 Test the hybrid-label-only explanation.**

**Why valid:** A label does not encode the adjacent acceptor or their relative orientation.

**Result:** Renaming hybridisation alone is insufficient; the alignment pathway supplies the mechanism.

**4 Separate model contribution from measured total energy.**

**Why valid:** Other energy terms and relaxation can also vary under torsion.

**Result:** No unique physical basis, numerical barrier or full total-energy law follows from this qualitative account.

# Fixed connectivity: $\text{H-C(=O)-NH}_2$ in bc

## End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

### Worked explanation

If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

**Follow the operation, its justification and its result.**

1. Relative donor/acceptor alignment is a model variable; donation magnitude needs additional physics.

2. Neither follows.
3. Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.

### **Result**

Neither follows.

### **Check**

Counterfactual: remove the word hybridisation while retaining changed orientation; the mechanistic statement survives.

**Mistake clinic • diagnose and repair**

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### **1-2-3 QUICK CHECK**

#### **1 • CHECK**

Counterfactual: remove the word hybridisation while retaining changed orientation; the mechanistic statement survives.

#### **2 • APPLY**

State which additional model would be needed for a numerical donation or barrier estimate.

### 3 • CONNECT

Relative donor/acceptor alignment is a model variable; donation magnitude needs additional physics.

3

## Execute the supplied cosine model and separately test the total-energy inference.

Hard concept

Section 3 of 10

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Why this section matters for the question set

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### 1 Evaluate the supplied overlap factor at the three angles.

**Why valid:** The stated model is  $\cos(\theta)$ ;  $\cos(0)=1$ ,  $\cos(60)=1/2$  and  $\cos(90)=0$ .

**Result:** Factors: 1, 1/2, 0.

### 2 The step the hard question turns on Identify the missing link from overlap to total energy.

**Why valid:** An overlap factor is not a Hamiltonian and supplies no full energy budget.

**Result:** The proposed total-energy conclusion is not warranted.

**3 Construct an explicit counterexample to uniqueness.**

**Why valid:** At fixed  $\theta=60^\circ$  an independently specified energy contribution can vary while the overlap factor stays  $1/2$ .

**Result:** Same overlap, different total energy; proportionality is not implied.

**4 Bound even alternative energy approximations.**

**Why valid:** A squared-interaction estimate needs stated weak-coupling assumptions; a degenerate two-level coupling can instead yield  $\epsilon \pm |t|$ .

**Result:** No universal cosine or cosine-squared total-energy law; both functions are even, so parity does not select one.

Fixed connectivity: H-C(=O)-NH2 in bc  
End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

**Keep alignment separate from total energy**

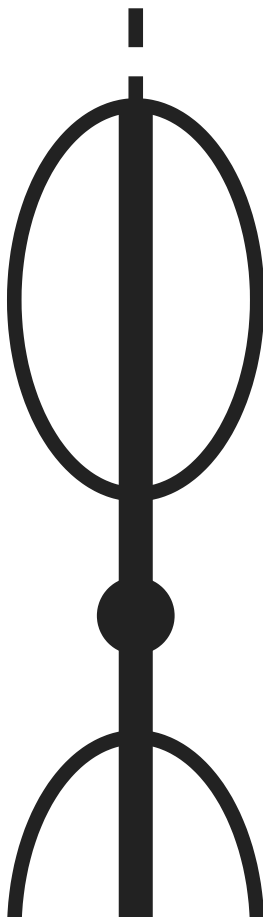
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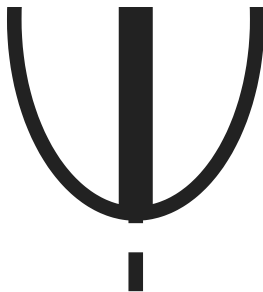
Relative angle  $\theta$  (degrees)



Independent toy energy contribution  $c$







# Axis points towards/away from viewer.

$\theta = 0^\circ$ ; given factor = 1; toy  $E = -1$ .

Declared demonstration only:  $g = \cos(\theta)$ ;  $E = -g^2 + c$  in arbitrary toy units. The square is an explicitly chosen example, not a universal law. Hold  $\theta$  fixed and vary  $c$ : the same  $g$  permits different  $E$ . Predict this before moving  $c$ . No physical barrier is computed.

## Worked explanation

If theta stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

**Follow the operation, its justification and its result.**

1. An overlap factor is not an energy Hamiltonian or the whole energy budget.
2. Overlap factor unchanged; total energy need not remain fixed.
3. Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling  $t_0 \cos(\theta)$  gives  $\epsilon \pm |t_0 \cos(\theta)|$ , so no universal cosine-squared total-energy law follows.

### Result

Overlap factor unchanged; total energy need not remain fixed.

### Check

At  $\theta=60$  degrees the factor is fixed at 1/2 while adding an independently specified energy contribution changes total energy. This falsifies uniqueness without modelling formamide quantitatively.

**Mistake clinic · diagnose and repair**

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### 1-2-3 QUICK CHECK

#### 1 · CHECK

At  $\theta=60$  degrees the factor is fixed at  $1/2$  while adding an independently specified energy contribution changes total energy. This falsifies uniqueness without modelling formamide quantitatively.

## 2 • APPLY

State assumptions needed for a weak-coupling second-order squared-interaction estimate.

## 3 • CONNECT

An overlap factor is not an energy Hamiltonian or the whole energy budget.

### Question-specific diagnosis and repair



### Now you do one

For a supplied direction factor  $g(\theta)=\cos(\theta)$ , compare  $30^\circ$  and  $60^\circ$ . Does the factor alone specify a total-energy difference? Hold  $60^\circ$  fixed and add a toy energy contribution  $c$ .

### Practice this concept in Core2

Q2

Q3

Q9

Sections +

## Delocalisation-sensitive hybridisation boundaries

**Subject:** Chemistry

**Topic:** Chemical Bonding and Molecular Structure

**Concept:** Delocalisation-sensitive hybridisation boundaries

**Concept difficulty:** Hard concept

1

**Resolve conflict between a local Lewis inventory and supplied aligned delocalisation.**

Hard concept

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### Conditions for this construction

- Can determine local Lewis structures and count electron domains.
- Understands resonance stabilization in introductory conjugated systems.

**Why this section matters for the question set**

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**1 Record the Lewis inventory and the supplied donation condition separately.**

**Why valid:** N has three bonds plus a lone pair, but the task also supplies approximately planar donation into

an adjacent carbonyl pi system.

**Result:** A four-entry Lewis ledger does not settle the electronic model.

**2 The step the hard question turns on Choose a basis that represents both jobs.**

**Why valid:** Three in-plane  $sp^2$ -like sigma directions can coexist with an aligned p-like lone-pair donor under the supplied model.

**Result:** A planar sigma framework plus a p-like donor connected to the carbonyl pi space.

**3 Explain why automatic  $sp^3$  assignment is inadequate.**

**Why valid:** A localized tetrahedral rule omits the stated delocalisation and alignment conditions.

**Result:** The local count is a useful ledger, not an overriding proof of the basis.

**4 Bound and counterfactually test the description.**

**Why valid:** Twisting or interrupting the donor-acceptor path changes the applicability; geometry alone does not uniquely measure hybrid coefficients.

**Result:** An introductory  $sp^2$ -like account is conditional, not unique observed orbital reality.

# Fixed connectivity: $\text{H-C(=O)-NH}_2$ in bc

## End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

### The key step

Approximately planar amide N can use an introductory  $\text{sp}^2$ -like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.

**Formula reference · meaning and conditions**

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### **Worked explanation**

Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?

### **Follow the operation, its justification and its result.**

1. A hybrid label describes a chosen basis; it does not independently cause donation.
2. No. Adjacency/alignment conditions differ. Repeating automatic four-domains-therefore-localized-sp3 in both exposes applicability failure.
3. Approximately planar amide N can use an introductory sp2-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.

### **Result**

No. Adjacency/alignment conditions differ. Repeating automatic four-domains-therefore-localized-sp3 in both exposes applicability failure.

## Check

Remove the hybridisation label: the alignment-plus-acceptor explanation must still describe the supplied donation condition.

## Mistake clinic • diagnose and repair



### 1-2-3 QUICK CHECK

#### 1 • CHECK

Remove the hybridisation label: the alignment-plus-acceptor explanation must still describe the supplied donation condition.

#### 2 • APPLY

Give one observation that constrains the model and one claim it cannot uniquely prove.

#### 3 • CONNECT

A hybrid label describes a chosen basis; it does not independently cause donation.

## 2 Test uniqueness, rather than merely finding one familiar angle.

Hard concept

Section 5 of 10

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Why this section matters for the question set

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### 1 Compare the supplied angles with their shared introductory label.

**Why valid:** CH<sub>4</sub>, NH<sub>3</sub> and H<sub>2</sub>O use sp<sup>3</sup>-like descriptions but have different lone-pair counts and nuclear shapes.

**Result:** 109.5, about 107 and about 104.5 degrees are compatible with the supplied shared label.

### 2 The step the hard question turns on Reject the unique-angle implication.

**Why valid:** A single model label does not fix all interactions that affect geometry.

**Result:** Angle deviations alone do not falsify every sp<sup>3</sup>-like description.

### 3 Include the amide boundary in the replacement rule.

**Why valid:** An aligned adjacent acceptor and delocalisation can make automatic localized four-domain classification inadequate.

**Result:** Local counts guide a bounded starting model; lone pairs, alignment and delocalisation require additional checks.

**4 Specify evidence needed for stronger physical claims.**

**Why valid:** Nuclear geometry alone does not uniquely recover wavefunctions or a hybrid decomposition.

**Result:** Use defined spectroscopic/electronic-structure evidence and model assumptions; retain alternative compatible descriptions.

Fixed connectivity:  $\text{H-C(=O)-NH}_2$  in bc  
End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

### **Worked explanation**

Can one four-domain label predict the same nuclear shape for CH<sub>4</sub> and H<sub>2</sub>O? Does observing a shape uniquely establish an orbital basis?

### **Follow the operation, its justification and its result.**

1. Several shapes/angles can share an introductory four-domain classification.
2. No to both.
3. CH<sub>4</sub>/NH<sub>3</sub>/H<sub>2</sub>O share common sp<sup>3</sup>-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.

### **Result**

No to both.

### **Check**

Independent counterexample: a shared four-domain label is compatible with 4,3,2 bonded-atom positions, so it cannot alone uniquely specify atom-only shape.

**Mistake clinic · diagnose and repair**

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Independent counterexample: a shared four-domain label is compatible with 4,3,2 bonded-atom positions, so it cannot alone uniquely specify atom-only shape.

#### 2 • APPLY

Name evidence needed before claiming a unique physical electronic description.

#### 3 • CONNECT

Several shapes/angles can share an introductory four-domain classification.

3

**Translate occupancy as well as geometry in the supplied radical model.**

**Hard concept**

Section 6 of 10

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**Why this section matters for the question set**

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**1 Read the supplied radical geometry and occupancy.**

**Why valid:** The benchmark specifies three planar C-H sigma directions and a perpendicular singly occupied p-like orbital.

**Result:** A three-direction sigma frame plus one unpaired electron.

**2 The step the hard question turns on Allocate the carbon valence basis.**

**Why valid:** An sp<sup>2</sup>-like sigma allocation uses three functions and leaves one p-like function perpendicular.

**Result:** Three hybrids plus one residual p-like function; four functions total.

**3 Place the supplied electron in the residual orbital.**

**Why valid:** CH<sub>3</sub> radical has 4+3=7 valence electrons; three sigma pairs use six.

**Result:** Residual p-like SOMO contains one electron, not zero.

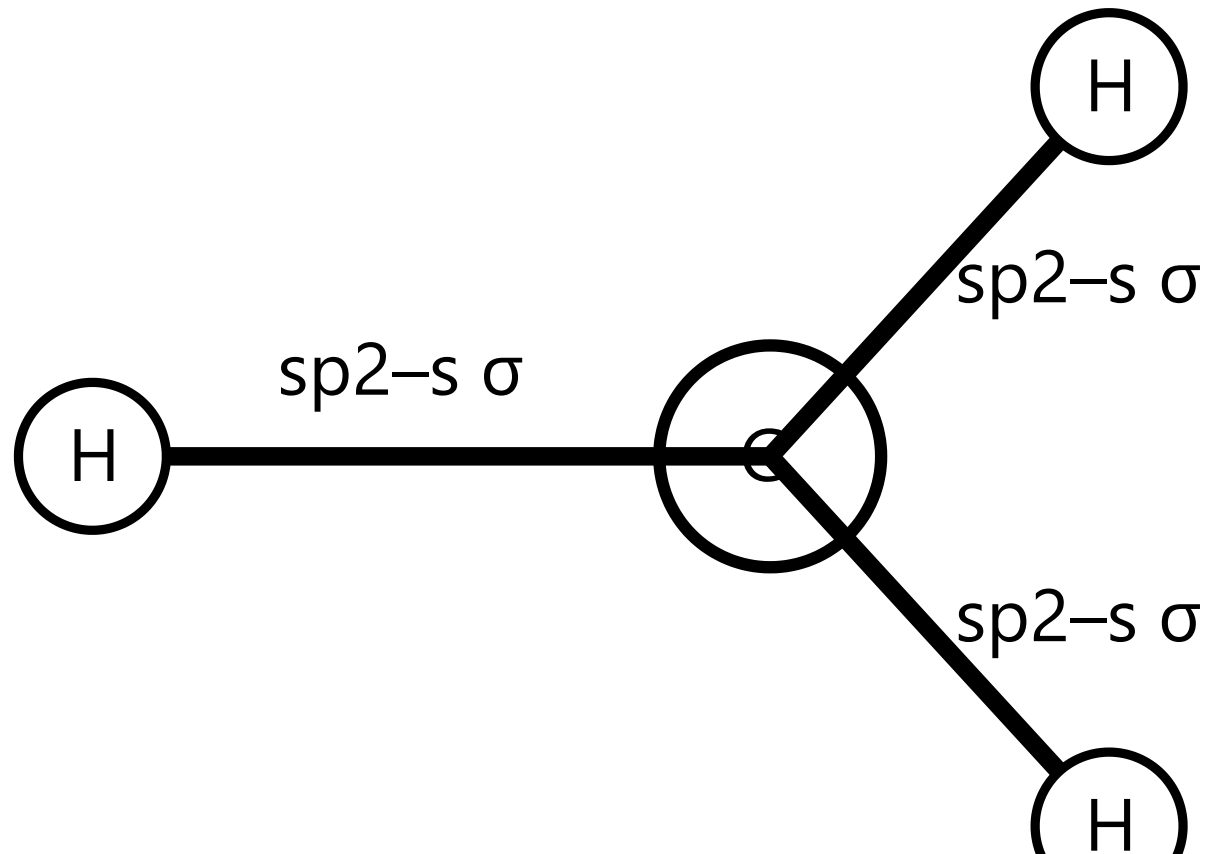
**4 Compare with a six-electron CH<sub>3</sub><sup>+</sup> model inventory.**

**Why valid:** Removing one electron changes residual occupancy while the three-bond count can remain.

**Result:** Three sigma bonds alone do not determine whether residual p is empty; the given radical model is not a universal claim.



# 1 · Planar $sp^2$ carbon sigma framework ( $120^\circ$ angl





Shows three C–H sigma bonds at  $120^\circ$  in the xy plane.

Stage 1 of 3

Staged orbital model of neutral methyl radical  $\text{CH}_3\cdot$ , showing the trigonal planar  $\text{sp}^2$  sigma framework and the perpendicular unhybridised p<sub>z</sub> orbital holding one unpaired electron (SOMO).

### Worked explanation

Compare  $\text{CH}_3$  radical (7 valence electrons) with a model  $\text{CH}_3^+$  inventory (6). Must their residual p occupancy be the same?

**Follow the operation, its justification and its result.**

1. A singly occupied orbital is not an empty orbital or a lone pair.
2. No: one versus zero in these supplied local models. Repeating empty p for both reveals bonds-only occupancy inference.
3. Supplied planar  $\text{CH}_3$  radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.

## Result

No: one versus zero in these supplied local models. Repeating empty p for both reveals bonds-only occupancy inference.

## Check

Independent count: CH<sub>3</sub> radical has  $4+3=7$  valence electrons; three sigma pairs use six, leaving one.

### Mistake clinic • diagnose and repair

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#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Independent count: CH<sub>3</sub> radical has  $4+3=7$  valence electrons; three sigma pairs use six, leaving one.

##### 2 • APPLY

Sketch the given radical while explicitly marking the singly occupied orbital.

##### 3 • CONNECT

A singly occupied orbital is not an empty orbital or a lone pair.

4

**Separate a discriminating observation from the strength of the inference it warrants.**

**Hard concept**

Section 7 of 10

[← Previous](#)[Next →](#)**Why this section matters for the question set**

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**1 State competing predictions before proposing observations.**

**Why valid:** The proposed localized pyramidal and donation-compatible planar accounts differ in nuclear arrangement and possible electronic consequences.

**Result:** Compare N geometry, relevant bond metrics and condition-specific electronic or dynamic signals.

**2 The step the hard question turns on Propose controlled geometry and bond measurements.**

**Why valid:** Defined species, phase and conditions allow predictions to be compared with uncertainty and alternative causes.

**Result:** Near-planarity can challenge a strictly pyramidal description; bond metrics add constraints.

**3 Add an independent electronic or torsional test.**

**Why valid:** A specified spectrum or calculation/experiment can test donation-related or rotation-related

predictions under stated assumptions.

**Result:** A second channel of evidence can discriminate accounts more strongly than geometry alone.

**4 Limit the warranted inference.**

**Why valid:** Compatible geometry can arise under multiple electronic descriptions; strength and barrier require quantitative evidence.

**Result:** No unique hybrid coefficients, donation magnitude or rotational barrier is measured by a qualitative planar sketch.

Fixed connectivity:  $\text{H-C(=O)-NH}_2$  in bc  
End-on view along C-N; centres projec

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

### **Worked explanation**

Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

### **Follow the operation, its justification and its result.**

1. A measurement can constrain multiple models without uniquely selecting an orbital decomposition.
2. No; a specified experiment or calculation and conditions/model are needed. Repeating direct geometry-to-strength inference exposes overinterpretation.
3. Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

### **Result**

No; a specified experiment or calculation and conditions/model are needed. Repeating direct geometry-to-strength inference exposes overinterpretation.

## Check

Use two compatible electronic descriptions of a planar structure as a counterexample to geometry-implies-unique-orbitals.

### Mistake clinic • diagnose and repair

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#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Use two compatible electronic descriptions of a planar structure as a counterexample to geometry-implies-unique-orbitals.

##### 2 • APPLY

Design a second independent observation and state one alternative explanation.

##### 3 • CONNECT

A measurement can constrain multiple models without uniquely selecting an orbital decomposition.

### Question-specific diagnosis and repair

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### Now you do one

A hypothetical p-like donor is supplied as aligned with a directly adjacent pi acceptor. Rotate its direction about the fixed connecting bond. What qualitative conclusion follows in this model, and what physical claim still needs evidence?

### Practice this concept in Core2

Q1

Q5

Q7

Q8

Sections+

## Local domain counting versus conjugated networks

**Subject:** Chemistry

**Topic:** Chemical Bonding and Molecular Structure

**Concept:** Local domain counting versus conjugated networks

**Concept difficulty:** Hard concept

### 1 Build a decision procedure that can fail its own applicability test.

Hard concept

## Conditions for this construction

- Understands orbital hybridisation in alkanes and alkenes.

### Why this section matters for the question set

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#### 1 Apply the same N Lewis ledger at both sites.

**Why valid:** Both neutral N sites have three sigma bonds and one lone pair.

**Result:** Local count alone does not distinguish A from B.

#### 2 The step the hard question turns on Trace donor-to-carbonyl adjacency and p availability.

**Why valid:** A has direct N-C(=O) adjacency; B contains the supplied saturated CH<sub>2</sub> interruption.

**Result:** A supports the stated contiguous donor/acceptor path; B lacks that ordinary path.

#### 3 Test alignment and validity before applying a label.

**Why valid:** Donation requires suitable orbital orientation as well as adjacency; spatial proximity does not create conjugation.

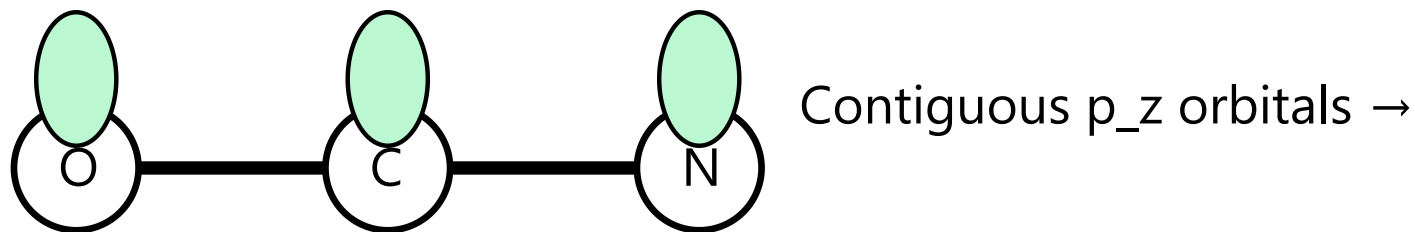
**Result:** Use conditional delocalised accounting at A and localized amine accounting at B within the given model.

#### 4 Construct a structural counterfactual.

**Why valid:** Replacing the saturated interruption with a p-bearing aligned bridge changes the path test.

**Result:** Reassess the model; absence of this ordinary path is not an absolute zero-interaction theorem.

### Site A · Direct amide: continuous aligned p-orbital highway



Shows adjacent sp<sup>2</sup> centers with aligned p<sub>z</sub> orbitals in formamide.

Stage 1 of 3

Comparison of Site A direct amide where contiguous p<sub>z</sub> orbitals enable delocalisation with Site B separated amine where intervening sp<sup>3</sup> CH<sub>2</sub> group blocks conjugation.

#### The key step

Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.

**Worked explanation**

Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?

**Follow the operation, its justification and its result.**

1. Local count and a continuous aligned pathway are independent checks.
2. No; spatial folding and orbital-connected adjacency are different changes.
3. Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.

**Result**

No; spatial folding and orbital-connected adjacency are different changes.

## Check

Changed case: replace a saturated interruption by a suitably aligned unsaturated bridge; reassess connectivity and orbital availability rather than reusing the answer.

### Mistake clinic • diagnose and repair



#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed case: replace a saturated interruption by a suitably aligned unsaturated bridge; reassess connectivity and orbital availability rather than reusing the answer.

##### 2 • APPLY

Specify a counterfactual bridge and test all procedure conditions again.

##### 3 • CONNECT

Local count and a continuous aligned pathway are independent checks.

## 2 Distinguish recalling a rule from establishing its applicability.

Hard concept

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Why this section matters for the question set

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### 1 Recall the local rule with its scope.

**Why valid:** Four domains classify ordinary localized introductory centres; recalling it does not show that locality holds.

**Result:** Starting ledger: three N bonds plus one lone pair.

### 2 The step the hard question turns on Inspect the supplied adjacency condition.

**Why valid:** A has a carbonyl acceptor directly adjacent; B has a saturated interruption in the ordinary p path.

**Result:** A needs a delocalisation/alignment test; B supports the local starting model here.

### 3 Identify evidence that defeats automatic application.

**Why valid:** Suitable donor/acceptor alignment, donation evidence or unusual occupancy add jobs the bond/pair total does not resolve.

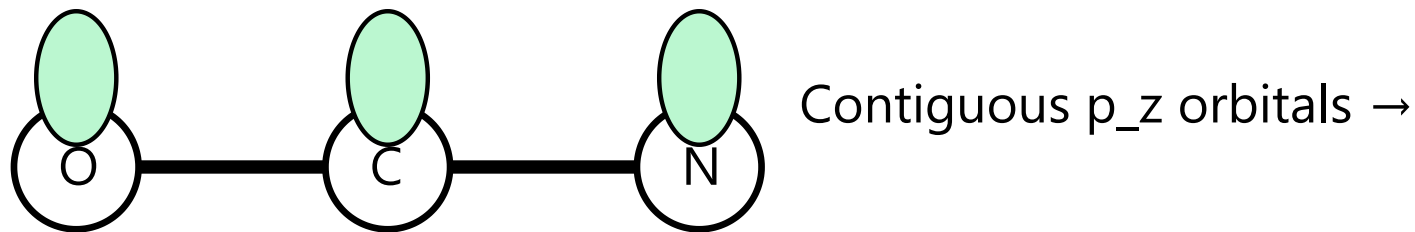
**Result:** Do not infer adequacy from the number four alone.

**4 Separate known conditions from unknown quantities.**

**Why valid:** Neither exact donor strength nor unique hybrid coefficients are supplied by this ledger.

**Result:** Apply a conditional classification or request the missing electronic evidence.

**Site A · Direct amide: continuous aligned p-orbital highway**



Shows adjacent sp<sup>2</sup> centers with aligned p<sub>z</sub> orbitals in formamide.

Stage 1 of 3

Comparison of Site A direct amide where continuous p<sub>z</sub> orbitals enable delocalisation with Site B separated amine where intervening sp<sup>3</sup> CH<sub>2</sub> group blocks conjugation.

**Worked explanation**

Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?

### **Follow the operation, its justification and its result.**

1. Local counting assumes the localized/domain model is useful for the asked task.
2. No; supplied adjacency/alignment evidence differs.
3. Four-domain counting is useful for localized introductory classification. Adjacent pi acceptors, aligned donor orbitals, planarity/donation evidence or radical occupancy can require a richer model.

### **Result**

No; supplied adjacency/alignment evidence differs.

### **Check**

Changed check: the same formal lone-pair entry at an amine and an amide can have different orbital-role descriptions under the supplied conditions.

**Mistake clinic • diagnose and repair**

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### **1-2-3 QUICK CHECK**

#### **1 • CHECK**

Changed check: the same formal lone-pair entry at an amine and an amide can have different orbital-role descriptions under the supplied conditions.

## 2 • APPLY

Separate a recalled count from a model-selection decision in a new conjugated case.

## 3 • CONNECT

Local counting assumes the localized/domain model is useful for the asked task.

### 3 Audit a proposed universal rule across cases that stress different boundaries.

Hard concept

Section 10 of 10

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#### Why this section matters for the question set

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#### 1 Audit each case for the explanatory job omitted by the slogan.

**Why valid:** NH<sub>3</sub> needs lone-pair versus nuclear-shape distinction; allene needs residual-p orientation; formamide needs donation conditions; the separated amine needs a pathway test.

**Result:** A label alone cannot supply all four explanations.

**2    The step the hard question turns on    Retain local counting as a starting ledger.**

**Why valid:** Atom-local sigma directions and central pairs give useful introductory information when localized assumptions hold.

**Result:** Count locally and conserve the valence basis.

**3    Add occupancy, adjacency and alignment branches.**

**Why valid:** A residual p function must have a specified occupancy and suitable neighbouring orbital path for the proposed interaction.

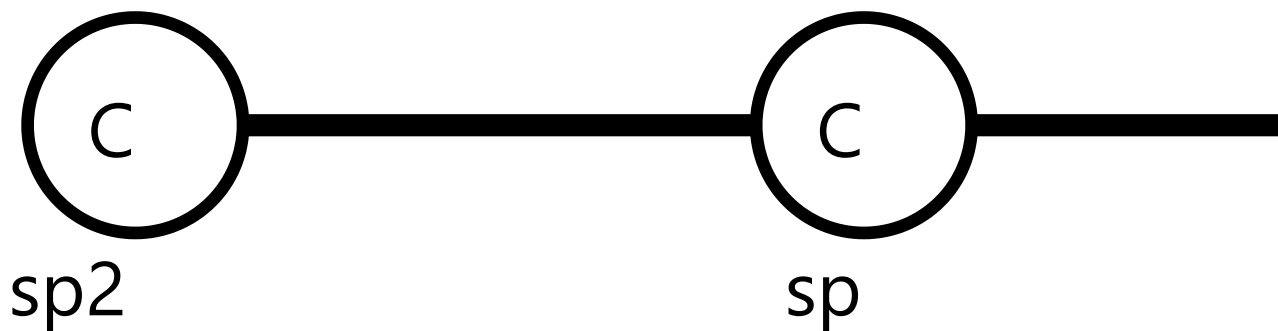
**Result:** Check what orbitals do, rather than treating the hybrid label as their cause.

**4    Include a validity and evidence stopping rule.**

**Why valid:** Quantitative geometry/reactivity or competing compatible electronic accounts require evidence beyond the ledger.

**Result:** Use a conditional framework and seek an explicit model or measurement when the simple rule is inadequate.

# 1 • Linear sigma axis and orbital hybridization (sp)



Shows linear C–C–C axis with sp<sup>2</sup>–sp–sp<sup>2</sup> hybridization.

Stage 1 of 4

Staged orbital model showing the central sp carbon of allene having two orthogonal p orbitals forming independent pi bonds with terminal sp<sup>2</sup> carbons, resulting in mutually perpendicular terminal CH<sub>2</sub> planes.

### **Worked explanation**

Would one bond/lone-pair count plus a label explain NH<sub>3</sub> shape, allene terminal-plane relation and amide donation by the same mechanism?

### **Follow the operation, its justification and its result.**

1. Counts, orbital roles, geometry and physical evidence are distinct layers.
2. No; their explanatory jobs differ. Repeating label-as-final-cause across all cases exposes an overgeneralised framework.
3. Use local count as a bounded starting model, then check residual p directions, occupancy, adjacency/alignment and evidence. A label alone explains neither every geometry nor all reactivity. NH<sub>3</sub>, allene and amide stress different limitations.

### **Result**

No; their explanatory jobs differ. Repeating label-as-final-cause across all cases exposes an overgeneralised framework.

## Check

Changed check: apply the framework to CH<sub>3</sub> radical; three bonds plus one unpaired electron require an occupancy step absent from the original slogan.

### Mistake clinic • diagnose and repair

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#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed check: apply the framework to CH<sub>3</sub> radical; three bonds plus one unpaired electron require an occupancy step absent from the original slogan.

##### 2 • APPLY

Produce a qualified procedure with an explicit stop-and-look-for-more-evidence branch.

##### 3 • CONNECT

Counts, orbital roles, geometry and physical evidence are distinct layers.

### Question-specific diagnosis and repair

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**Now you do one**

An introductory local model proposes a direct aligned donor/acceptor path. Insert a saturated spacer. State the local-model effect and one claim it does not establish.

**Practice this concept in Core2**

Q4

Q6

Q10